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RESEARCH ARTICLE

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Explainable multi-modal Vision Transformer fusion for malware classification

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ABSTRACT

The rapid evolution of malware variants has significantly intensified cybersecurity threats, posing critical challenges for traditional signature-based detection and mitigation approaches. To address the limitations of traditional approaches, our paper proposes a Cross-Attention-Based Multi-modal Vision Transformer for malware detection and classification using both malware image representations and behavioral features. A Vision Transformer (ViT) is employed to capture spatial malware image characteristics, simultaneously extracting behavioral network features and learning semantic representations. The integration is effectively done using cross-attention fusion and is utilized for multi-class malware classification across the malware families. The experimental evaluation achieved an overall classification accuracy of 88%, outperforming malware fusion-based approaches.

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Malware imaging; Vision Transformer; cross-attention multi-model fusion; explainability analysis; LLM-based threat reporting

1 Introduction

The increasing rate of malware attacks poses significant challenges to information and communication systems. Modern malware variants employ obfuscation, polymorphism and behavioral evasion techniques to bypass traditional Intrusion Detection Systems (IDS) [1]. Meanwhile, machine learning and deep learning approaches have gained attention in the recent years for identification and mitigation of cyber attacks [2]. One of the popular Malware visualization techniques is to transform malware binaries into grayscale images [3], enabling computer vision models to identify malware-specific texture and structural patterns. Convolutional Neural Networks (CNNs) [4] and Vision Transformers (ViTs) [5] have shown promising performance with visual malware features. However, image-only approaches often fail to capture runtime behavioral features to identify sophisticated malware families. Behavioral analysis, such as API calls, registry activities, network behavior and file operations, provides more detailed information about the malware during its execution. Integrating visual and behavioral representations improves the robustness of malware detection and generalization capability. This paper proposes a Cross-Attention-Based Multi-modal Vision Transformer approach that combines malware image representations along with behavioral feature representations, learning the semantic relationships between them, thereby improving malware classification performance. Our contributions are as follows:

1. An attention-guided multi-modal feature representation is developed by integrating adaptive image encoding, semantic-aware and behavioral-guided malware.
2. A Vision Transformer (ViT) with 16 × 16 patch embedding is employed to extract robust visual features from malware images.
3. A multimodal cross-attention fusion is proposed to combine features into a 768-dimensional attended representation for accurate malware classification and family prediction.

The paper is organized as follows: Section 2 provides the related work. Section 3 provides the Methodology. The results and discussion are given in Section 4 and the conclusion in Section 5.

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2 Related work

Malware detection and classification are progressing from traditional signature based methods to modern systems leveraging machine learning and deep learning techniques [1,6]. Traditional approaches often struggle to detect emerging and previously unseen malware variants [3], this motivates the development of data-driven malware detection. Recent research has increasingly focused on the integration of Vision transformer-based architectures, multi-modal feature learning, explainable artificial intelligence (XAI) and Large Language Models (LLMs) [5,7]. One of the first studies on malware visualization and Image-Based analysis was conducted by Nataraj et al. [3], here transformation of malware binaries into visual representations is an effective approach for malware classification. Convolutional Neural Networks (CNNs) are extensively used in malware image classification due to their ability to extract hierarchical visual features [4]. Edmar et al. [8] demonstrated the use of ResNet for malware image classification with improved feature learning capabilities. However, CNN-based approaches mainly rely on local receptive fields.

Deep learning methods are being replaced by transformer-based architectures employing attention mechanisms. Vaswani et al. [9] proposed transformer-based self-attention mechanisms that capture global contextual relationships. Dosovitskiy et al. [5] introduced Vision Transformer (ViT), unlike CNNs, ViT, with the help of self-attention mechanisms capture long-range spatial dependencies as well as global image context, making ViT suitable for malware analysis.

Akbari et al. [10] proposed a transformer-based architecture for multi-modal representation learning across different domains such as visual, audio and textual. The reported results demonstrate the effectiveness of fusion mechanisms in learning from diverse inputs. Similarly, Tsai et al. [7] demonstrated attention-based fusion on different sources preserving source-specific information. Selvaraju et al. [11] introduced Grad-CAM, a widely adopted model by CNN, a gradient-based visualization technique for model predictions. Lundberg and Lee [12] proposed SHapley Additive exPlanations (SHAP), to understand feature contributions that facilitate transparent malware reasoning. Loshchilov and Hutter [6] introduced AdamW, an efficient optimization algorithm in deep neural networks that decouples weight decay from gradient updates.

Large Language Models (LLMs) are recently used in cybersecurity applications. Sarhan et al. [13] presented LLM applications in cybersecurity, highlighting their uses such as threat intelligence, malware analysis, vulnerability assessment and security automation. Similarly, Hamal et al. [14] emphasized the potential for improving analyst productivity and decision support using LLMs. Yue et al. [15] discussed the opportunities and challenges associated with deploying LLMs in security environments. Recently Khan et al. [16] provided a review of LLM-based techniques for cyber threat intelligence through automated reasoning. Integrating LLMs with Multi-modal systems bridges the gap by transforming model outputs into interpretable threat reports.

Seneviratne et al. [17] proposed SHERLOCK, a self-supervised Vision Transformer work designed for binary malware detection from image-based binary representations. SHERLOCK achieves an accuracy of 97% for binary malware detection using approximately 1.2 million Android applications collected from the AndroZoo repository. However, we have not compared SHERLOCK's accuracy as a direct and controlled comparison with our results because the datasets, the objectives and the experimentation procedures are different in both cases. In particular, our work focuses on multimodal malware family classification.

In summary, unlike traditional signature-based techniques and advanced deep learning as well as transformer-based approaches, which focus on malware visualization and CNN-based malware classification, recent works have explored more on Vision Transformers, multi-modal feature extraction and feature fusion, explainable artificial intelligence (XAI) and large language models (LLMs) to improve malware detection accuracy. Although these approaches have promising results, most existing methods either rely solely on visual representations or lack effective integration of behavioral feature information and explainable threat reasoning. To address these limitations, our work introduces an attention-guided multi-modal malware analysis combining both malware image representations with malware behavioral features through a cross-attention fusion mechanism. Also, using explainable AI and LLM-driven threat reporting to provide more interpretable malware family predictions along with the detection accuracy, achieving an overall accuracy of 88%, highlighting the effectiveness of multimodal feature learning and attention-based fusion for malware detection and analysis.

3 Methodology

This section presents our proposed architecture. Initially the raw executable files consisting of static features such as entropy, opcode frequency, APIs and behavioral features such as API calls, registry activities, network traffic etc, are converted into grayscale image representation suitable for deep learning and Large Language Models (LLMs)-based analysis. Traditional malware analysis approaches often struggle to generalize evolving malware, recent approaches preserve both static and dynamic characteristics of the malicious code, supporting multiple image-generation strategies. Malware binaries are converted into grayscale images, each malware sample is represented as a sequence of unsigned 8-bit integer values corresponding to its raw binary content [3]. Let the one dimensional (1D) malware binary be represented as:

$$X = x_1, x_2, x_3, \ldots, x_n, \tag{1}$$

where $x_i \in [0, 255]$ denotes the ith byte value of the malware binary and n represents the total number of bytes.

A sequence of 8-bit unsigned integers (bytes), with each value ranging from 0 to 255, is mapped to a grey scale pixel and subsequently reshaped into a two-dimensional (2D) image [3]. The generated images are the visual encoding of the 1D numerical feature vector to a 2D grayscale representation.

To obtain a consistent pixel representation, the byte sequence is normalized using min-max normalization:

$$X_{\text{norm}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}, \tag{2}$$

where X_{min} and X_{max} denote the minimum and maximum byte values. Later, the normalized sequence is reshaped into two-dimensional grayscale image

$$I \in \mathbb{R}^{h \times w}, \tag{3}$$

where h and w represent the height and width of the generated malware image. All important structural patterns, texture information and local dependencies within malware samples are retained. The RGB images capture additional semantic information such as entropy levels, the opcode distribution and different section-specific characteristics into separate color channels. The entropy-based heatmaps highlight the area of importance, which exhibits abnormal randomness associated with obfuscation, encryption, or packaging techniques. Beyond these methods, we introduce:

Adaptive image encoding: This employs a dynamic encoding strategy where the image generation process depends on factors such as binary composition, file structure, entropy distributions and extracted behavioral features. This produces more informative visual representations than compared to fixed grayscale mappings.

Semantic-aware malware visualization: Malware executable sections, API calls, opcode sequences and metadata are used into the image-generation process. They help in understanding of malware structures and its functionalities.

Behavioral-guided image construction: Dynamic features, including API call patterns, network activities, file-system interactions, are used for visual representation.

By combining these, we create rich feature spaces for malware classification, explainability and LLM-based reasoning. Figure 1 illustrates the overall proposed architecture. In the proposed methodology the CSV feature records provided by the CCCS-CIC-AndMal2020 Android malware dataset are preprocessed, later, the resulting numerical feature vectors are transformed into fixed-size 2D grayscale images for the visual representations. These representations are further processed by a Vision Transformer. We clearly emphasize that the generated malware images are the feature-to-image encodings and therefore should not be interpreted as the direct spatial representations of the APK/DEX byte sequences. In parallel, 141 behavioral features are also extracted, with non-numeric attributes removed, by handling missing values. A feature normalization is carried out before projecting into a 128-dimensional behavioral representation. On the other hand, the 224×224 images are divided into 16×16 patches, and processed through the Transformer encoder by obtaining the class-token visual

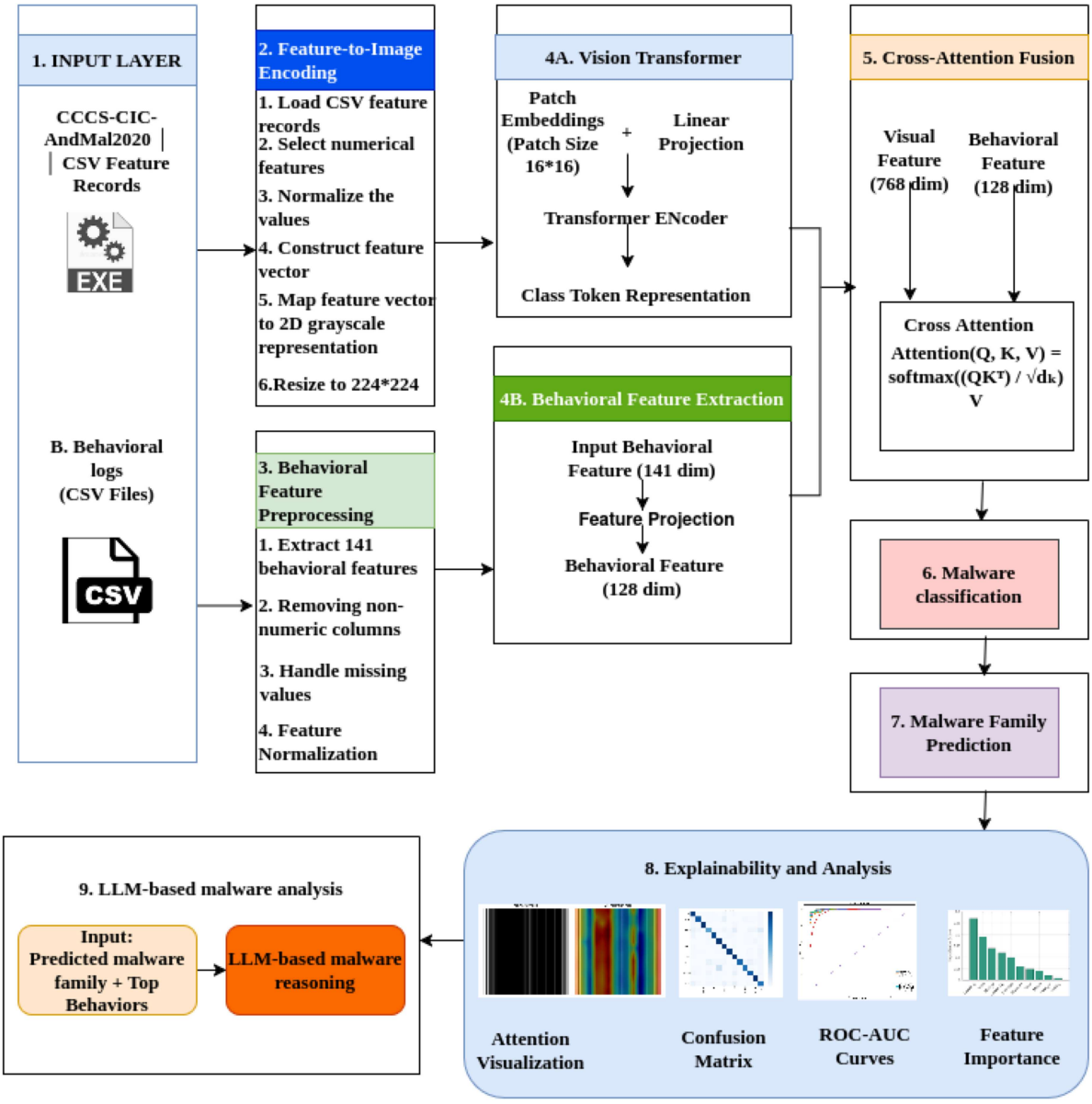


Figure 1. Overall architecture of the proposed work.

representation. The resultant 768-dimensional visual representation and 128-dimensional behavioral representation are integrated through a cross-attention fusion mechanism. This fusion is used for malware classification and malware-family prediction. Finally, LLM-based malware analysis is done with rigorous evaluation, which provides higher-level reasoning.

3.1 Behavioral feature extraction and feature encoder

In addition to visual, behavioral and network-based features are extracted to capture characteristics that provide insights beyond image-based analysis. Let the behavioral feature vector be represented as,

$$B = b_1, b_2, b_3, \ldots, b_m, \tag{4}$$

where b_i denotes the ith behavioral attribute and m represents the total number of extracted features. They have numerical ranges, hence feature standardization is done using z-score normalization.

$$B_{\text{norm}} = \frac{B - \mu}{\sigma}, \tag{5}$$

where μ denotes the mean value of the feature and σ is standard deviation.

Later for compact representation, they are processed further processed into a behavioral feature encoder with fully connected neural network interleaved with Rectified Linear Unit (ReLU) activation functions. It learns a latent representation from the input features vector:

$$F_{\text{beh}} = f(B), \tag{6}$$

where $f(\cdot)$ denotes the behavioral encoding network and F_{beh} is resulting behavioral feature embedding. The final behavioral embedding captures salient behavioral patterns associated with malware activities.

$$F_{\text{beh}} \in \mathbb{R}^{d_b}. \tag{7}$$

3.2 Vision Transformer (ViT)-based malware feature extraction

Unlike conventional Convolutional Neural Networks (CNNs), ViT captures long-range dependencies and global contextual information through self-attention mechanisms. Each malware image is divided into fixed-size image patches (P_i represents the image patch). The embedded patches are processed through multiple transformer encoder layers employing multi-head self-attention. Figure 2 demonstrates model interpretability by extracting inter ViT feature tokens. In Figure 2, the red regions indicate the image patches contributing strongly to the malware classification decision.

The attention operation is defined as,

$$I \rightarrow P_1, P_2, \ldots, P_n, \tag{8}$$

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \tag{9}$$

where Q, K and V represent the query, key and value matrices and d_k is the dimensionality of the key vectors. The self-attention mechanism enables the model to learn interactions between image patches by assigning adaptive importance weights to different regions of the malware image. Here we get high-level feature representation providing good description of malware image behavior:

$$F_{\text{ViT}} \in \mathbb{R}^{d_v}, \tag{10}$$

where d_v denotes the dimensionality of the learned feature space.

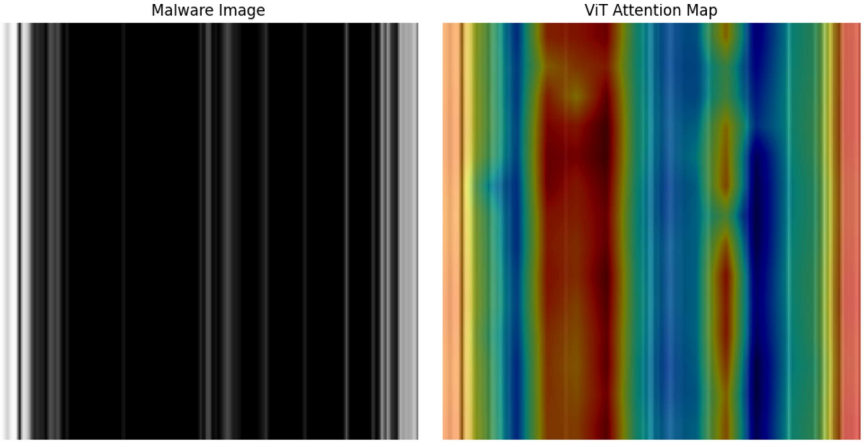


Figure 2. Attention visualization of malware image.

3.3 Behavioral feature fusion and classification

Further, the extracted ViT features and behavioral embeddings are unified into multi-model. This multi-model captures a broader view of malware behavior. A Cross-Attention-Based transformer architecture is used to learn the relationships between visual patterns and behavioral features, identifying the most relevant features instead of considering all the features directly. The Vision Transformer (ViT) extracts 768D visual features, while the cross-attention module produces 128D attention-enhanced behavioral features. These two feature vectors together form a 896D fused representation, which serves as the input to the final classification network. This fusion will help in collecting the behavioral evidence during execution. With the final fused feature representation (Ffused), the classifier estimates the probability distribution over malware classes using the Softmax function. Let the visual feature representation extracted from the ViT, the behavioral feature representation, cross-attention fusion and softmax function be denoted as follows:

$$F_{\text{img}} \in \mathbb{R}^{d_i}, \tag{11}$$

$$F_{\text{beh}} \in \mathbb{R}^{d_b}, \tag{12}$$

$$F_{\text{fusion}} = \text{CrossAttention}(F_{\text{img}}, F_{\text{beh}}). \tag{13}$$

where F_{fusion} represents the integrated feature embedding.

$$\hat{Y} = \text{Softmax}(WF_{\text{fused}} + b), \tag{14}$$

where W denotes the trainable weight matrix, b represents the bias vector and $\hat{Y}$ corresponds to the predicted class probability distribution. To further optimize, categorical cross-entropy loss is used:

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i), \tag{15}$$

where y_i denotes the ground-truth class label, $\hat{y}_i$ represents the predicted probability for class i and C is number of malware classes. The network is trained using the AdamW optimization algorithm.

3.4 Attention-guided explainability

Unlike post hoc explanations, we see better transparency in transformer attention weights identifying the regions of malware images contributing most to the model's predictions and is shown in Figure 2. To better understand the rationale, visual evidence, including self-attention maps, patch-level importance and token-based explanation scores are collected. The transformer attention operation:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

where (Q), (K) and (V) denote the query, key and value matrices and (d_k) represents the dimensionality of the key vectors.

3.5 LLM-generated malware reasoning and SHAP-based feature importance analysis

Human-readable explanations are very important and they influence the decisions based on the information, helping LLM in contextual reasoning. It translates technical explanations into structured explanations by analyzing sources. Along with producing only a malware label, the LLM explains why a sample exhibits specific malicious characteristics and highlights the evidence. On the other side SHapley Additive exPlanations (SHAP) is used to understand the contribution of behavioral attributes. It assigns an importance score to each feature based on its influence on the prediction outcome. The SHAP value for feature (i) is computed and shown in Figure 3.

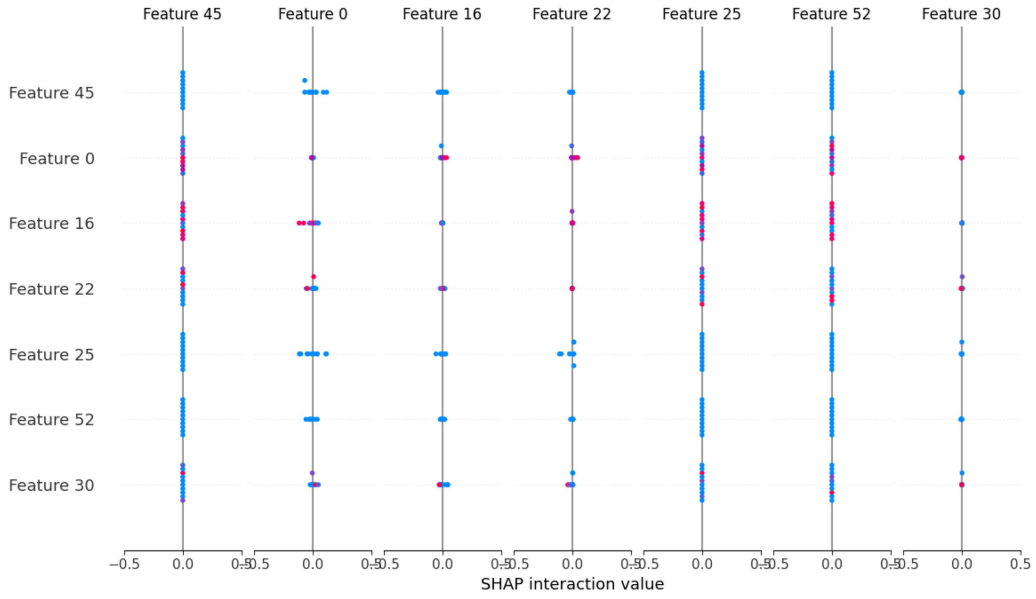


Figure 3. SHAP values.

$$\phi_i = \sum_{S \subseteq F \setminus i} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup i) - f(S)], \tag{16}$$

where ϕ_i denotes the contribution of feature i, F represents the complete feature set and S denotes a subset of features.

4 Results and discussion

Although traditional machine learning models may achieve competitive results, our work has additional advantages through Multi-model learning, attention-based explainability and LLM-based threat reasoning, achieving accuracy of 88% on 14 different malware families. The discriminative capability of the model is improved by integrating the image representations and the behavioral features. During the training process, the loss value decreases from 3012.54 to 831.08 over ten epochs, indicating effective optimization and robust malware feature learning.

This work utilizes the CCCS-CIC-AndMal2020 [18] dataset divided into training and testing subsets using stratified sampling, consisting of approximately 400,000 Android applications. These records describe various aspects of application behavior, including communication activities, traffic statistics, protocol-level information and other execution-related characteristics. The model is trained using the AdamW optimizer with a learning rate of 1×10^{-4} and a batch size of 16, with categorical cross-entropy loss for 10 training epochs. All experiments are conducted on an NVIDIA RTX A6000 GPU-based Mini-Batch Learning using CUDA acceleration. A checkpoint mechanism is implemented that periodically saves model parameters and resumes training automatically after interruptions.

4.1 Baseline experimental results

The baseline experiments were evaluated using the common test set. The results show that the Random Forest and Behavioral-only MLP models provide strong classification performance, while the ViT-only model provides the image-based baseline. The performance varies across different malware families, particularly for classes with small support, resulting in relatively lower macro F1-scores compared with weighted F1-scores.

Initially, a controlled experiment was performed using random seed 42. Since a single run with random seed 42 does not provide sufficient information to calculate the standard deviation, five independent runs

were subsequently conducted using the same experimental setup and the random seeds 42, 123, 2024, 3407 and 5678. The mean and standard deviation were then calculated across these five runs.

Accuracy, weighted precision, weighted recall, weighted F1-score, macro precision, macro recall, macro F1-score and weighted multiclass ROC-AUC were calculated for each run. Tables 1–5 present the seed wise results on the common test set for the following baseline models: Random Forest, 1D-CNN, ViT-only, Behavioral-only MLP and Bi_LSTM. The individual seed-wise results are reported in the corresponding baseline tables, while the overall five-seed comparison is presented using mean ± standard deviation.

The five-seed results provide a more robust assessment of model performance than a single random run. The mean ± standard deviation results are as follows:

- **Random forest:** accuracy 92.66 ± 0.08%, weighted F1 92.35 ± 0.08%, macro F1 90.27 ± 0.15% and ROC-AUC 99.62 ± 0.01%.

Table 1. Random forest baseline results on the common test set using random seed 42, 123, 2024, 3407, 5678.

Seed	Accuracy (%)	Weighted precision (%)	Weighted recall (%)	Weighted F1 (%)	Macro precision (%)	Macro recall (%)	Macro F1 (%)	ROC-AUC (%)
42	92.7957	93.0633	92.7957	92.5000	95.2719	87.2524	90.4715	99.6258
123	92.6553	92.9465	92.6553	92.3468	95.0853	86.9409	90.2429	99.6316
2024	92.5992	92.8764	92.5992	92.2928	94.5975	86.9987	90.0595	99.6227
3407	92.6366	92.8898	92.6366	92.3182	94.9602	87.0474	90.2219	99.6095
5678	92.6272	92.9370	92.6272	92.3097	95.0795	87.1963	90.3468	99.6210

Table 2. 1D-CNN baseline results on the common test set using random seed 42, 123, 2024, 3407, 5678.

Seed	Accuracy (%)	Weighted precision (%)	Weighted recall (%)	Weighted F1 (%)	Macro precision (%)	Macro recall (%)	Macro F1 (%)	ROC-AUC (%)
42	26.3005	6.9172	26.3005	10.9535	1.8786	7.1429	2.9748	75.1022
123	30.2395	39.7843	30.2395	29.4562	16.5254	17.6454	13.8781	76.1961
2024	26.3005	6.9172	26.3005	10.9535	1.8786	7.1429	2.9748	60.6370
3407	29.9869	19.8062	29.9869	19.4549	9.3298	11.3746	6.3424	79.8643
5678	6.3810	0.4740	6.3810	0.8825	0.5306	7.1429	0.9878	62.7158

Table 3. ViT-only baseline results on the common test set using random seed 42, 123, 2024, 3407, 5678.

Seed	Accuracy (%)	Weighted precision (%)	Weighted recall (%)	Weighted F1 (%)	Macro precision (%)	Macro recall (%)	Macro F1 (%)	ROC-AUC (%)
42	94.7043	95.0020	94.7043	94.4526	95.3314	91.6107	93.0627	99.5831
123	95.2564	95.5347	95.2564	95.0357	94.3446	94.0751	93.8117	99.5677
2024	94.7231	94.9969	94.7231	94.4262	94.9196	92.8780	93.4113	99.7054
3407	94.6388	94.7885	94.6388	94.3583	95.8593	92.5790	93.8567	99.6529
5678	94.9663	95.0884	94.9663	94.7160	95.3726	92.7993	93.7720	99.6578

Table 4. Behavioral-only MLP baseline results on the common test set using random seed 42, 123, 2024, 3407, 5678.

Seed	Accuracy (%)	Weighted precision (%)	Weighted recall (%)	Weighted F1 (%)	Macro precision (%)	Macro recall (%)	Macro F1 (%)	ROC-AUC (%)
42	96.2668	96.4881	96.2668	96.0181	98.2044	95.7605	96.6766	99.5863
123	96.2668	96.4898	96.2668	96.0179	98.0181	95.7670	96.5804	99.6389
2024	96.2575	96.4696	96.2575	96.0063	98.7058	95.7068	96.9188	99.4869
3407	96.2481	96.4411	96.2481	95.9894	98.5324	95.6703	96.8064	99.6997
5678	96.2856	96.4836	96.2856	96.0285	98.5674	95.7292	96.8560	99.4448

Table 5. Bi-LSTM baseline results on the common test set using random seed 42, 123, 2024, 3407, 5678.

Seed	Accuracy (%)	Weighted precision (%)	Weighted recall (%)	Weighted F1 (%)	Macro precision (%)	Macro recall (%)	Macro F1 (%)	ROC-AUC (%)
42	22.9900	22.5900	22.9900	20.5000	14.6900	12.8300	11.5300	59.3500
123	33.5423	38.0927	33.5423	29.0966	21.0263	15.9449	14.6323	61.0858
2024	52.8817	57.4482	52.8817	48.1815	32.0646	23.0157	22.4933	68.2385
3407	37.0696	44.6172	37.0696	34.2136	25.8629	19.4541	18.7503	65.5551
5678	40.5408	38.6219	40.5408	36.7987	17.1388	17.6979	16.2718	66.0254

- **1D-CNN:** accuracy 23.84 ± 9.95%, weighted F1 14.34 ± 10.71%, macro F1 5.43 ± 5.10% and ROC-AUC 70.90 ± 8.64%.
- **ViT-only:** accuracy 94.86 ± 0.26%, weighted F1 94.60 ± 0.28%, macro F1 93.58 ± 0.34% and ROC-AUC 99.63 ± 0.06%.
- **Behavioral-only MLP:** accuracy 96.26 ± 0.01%, weighted F1 96.01 ± 0.01%, macro F1 96.77 ± 0.14% and ROC-AUC 99.57 ± 0.11%.
- **Bi-LSTM:** accuracy 37.40 ± 10.86%, weighted F1 33.76 ± 10.19%, macro F1 16.74 ± 4.15% and ROC-AUC 64.05 ± 3.69%.

The overall five-seed comparison of all baseline models is presented in Table 6. These controlled experiments provide a common evaluation setting for comparing the visual-only, behavioral-only and sequence-based baseline approaches with the proposed multimodal architecture.

4.2 Training and classification results

The training process demonstrated stable convergence, corresponding to a reduction of approximately 72.4%. This decline in the objective function indicates effective optimization of the model learning discriminative malware characteristics from both visual and behavioral representations. The overall classification accuracy is 88% on 10,688 testing samples. The weighted F1-score is 88%. Furthermore, the macro-averaged performance metrics reflect the robustness in handling class imbalance and maintain prediction performance across malware categories.

4.2.1 Class-wise performance and behavioral feature importance analysis

The classification report say Riskware achieved the highest classification effectiveness with a precision of 0.94, recall of 0.94 and F1-score of 0.94. Similarly, Trojan_Spy and Adware obtained F1-scores of 0.94 and 0.91, respectively and shown in Table 11. Performance variations were observed in minority classes, but the model maintained competitive recall values, indicating effective detection. Feature importance analysis with behavioral feature encoder identified a subset of highly influential behavioral features that contributed significantly to the malware classification and is shown in Table 7. Among them, Feature 73 exhibited the highest importance score of 7.9384. With these advances, multi-modal architectures and cross-attention mechanisms gained attention.

Table 6. Overall five-seed comparison of baseline models on the common test set.

Baseline	Accuracy (%)	Weighted F1 (%)	Macro F1 (%)	ROC-AUC (%)
Random forest	92.66 ± 0.08	92.35 ± 0.08	90.27 ± 0.15	99.62 ± 0.01
1D-CNN	23.84 ± 9.95	14.34 ± 10.71	5.43 ± 5.10	70.90 ± 8.64
ViT-only	94.86 ± 0.26	94.60 ± 0.28	93.58 ± 0.34	99.63 ± 0.06
Behavioral-only MLP	96.26 ± 0.01	96.01 ± 0.01	96.77 ± 0.14	99.57 ± 0.11
Bi-LSTM	37.40 ± 10.86	33.76 ± 10.19	16.74 ± 4.15	64.05 ± 3.69

Table 7. Top 10 behavioral features based on importance scores.

Feature index	Importance score
73	7.9384
71	1.5047
44	1.0032
132	0.9189
136	0.9090
80	0.7795
67	0.7588
4	0.7276
135	0.6833
42	0.6578

4.2.2 LLM-based malware reasoning

To support rigorous evaluation, a stratified subset of 669 cases covering 14 different malware classes is constructed. Here, each explanation is assessed using prediction fidelity, grounding of evidence and features, interpretation and hallucination risk. The implementation is in 2 phases. Before large-scale generation, we conducted a 10-case pilot using the Qwen:8B model and details of the 10 case-level results are given in Table 9. We received valid explanations for all 10 cases with no generation failures. The overall mean explanation score is 86.25/100, and the mean feature-grounding ratio is 0,720, prediction fidelity is 4.00/4, evidence grounding is 3.60/4, interpretation quality is 3.60/4 and limitation awareness is 2.60/4. Hallucination risk is categorized as low, medium, or high. Every evaluated case is retained for human review. The screening indicators and hallucination-risk screen flagged 1 case as low risk, 5 as medium risk and 4 as high risk. It also included an incorrectly classified case: CASE_02525 (Backdoor(true class) is ViT-predicted as Riskware with confidence 0.9856), demonstrating that the explanation must describe the model prediction, and it must not convert the model confidence into evidence to say the actual malware behavior. The LLM explanation quality averages reported are from the 669-case generation. Table 8 presents the Quantitative evaluation of LLM generated explanations. They assess the reliability of the generated explanations for the multimodal ViT predictions. These cases are also retained for human review. The LLM model is explicitly told not to invent any network activity, APIs, permissions, data theft, SMS behavior or any other malware behavior unless it's supported by supplied evidence. The generation process is successful for all 669 cases. 10 case-level results of LLM-generated explanations are given in Table 9. Table 10 gives 669-case LLM generation report.

The generation process is successful for all 669 cases. In that, 492/669 are classified as valid, and the remaining ones are invalid.

4.2.3 Sample LLM report

Unlike generating only a malware label, the model also produces contextual explanations of the observed behavior as follows:

Table 8. Quantitative evaluation of LLM-generated explanations.

Metric	Result	What it measures
No. of pilot cases	669	Quantitative assessment
Prediction-correct cases	501 (74.89%)	Cases where the ViT prediction matched the ground-truth label
Prediction-incorrect cases	168 (25.11%)	Cases where the ViT prediction differed from the ground-truth label; these assess whether the LLM explains incorrect predictions without falsely presenting them as correct
Mean prediction score	4.000/4	Correct reporting of model prediction, confidence and correctness
Mean prediction confidence	0.7853	Average confidence associated with the model predictions
Confidence SD	0.2258	Variation in prediction confidence across the 669 cases
Confidence range	0.1486–1.0000	Minimum–maximum prediction confidence
Mean evidence score	3.600/4	Use of the supplied behavioral features in the explanation
Mean interpretation score	3.600/4	Separation of evidence, interpretation and unsupported assumptions
Mean limitation score	2.600/4	Whether the explanation states what cannot be concluded
Mean overall explanation score	86.25/100	Overall quality of the generated explanations across the evaluated dimensions
Mean feature grounding ratio	72.0%	Degree to which explanation content refers to supplied features
Low hallucination risk	1/10 (10%)	Conservative screening for potentially unsupported claims
Medium hallucination risk	5/10 (50%)	Conservative screening for potentially unsupported claims
High hallucination risk	4/10 (40%)	Conservative screening for potentially unsupported claims

Table 9. 10 case-level results of LLM-generated explanations.

Case	True	Predicted	Correctness	Prediction score	Evidence score	Interpretation score	Limitation score	Overall explanation score	Risk
CASE_04695	Scareware	Scareware	Yes	4	3	3	2	75.00	High
CASE_01022	No_Category	No_Category	Yes	4	4	4	2	87.50	High
CASE_02436	Trojan_Dropper	Trojan_Dropper	Yes	4	4	3	4	93.75	Medium
CASE_10540	Trojan_SMS	Trojan_SMS	Yes	4	3	3	4	87.50	Medium
CASE_09512	Ransomware	Ransomware	Yes	4	3	4	2	81.25	Medium
CASE_00454	Scareware	Scareware	Yes	4	3	4	2	81.25	Low
CASE_04660	Riskware	Riskware	Yes	4	4	4	2	87.50	Medium
CASE_09990	PUA	PUA	Yes	4	4	3	2	81.25	High
CASE_05442	Trojan_SMS	Trojan_SMS	Yes	4	4	4	4	100.00	High
CASE_02525	Backdoor	Riskware	No	4	4	4	2	87.50	Medium

Table 10. 669-case LLM generation report.

Metric	Result
Total number of cases	669
Unique case IDs	669
Successful generations	669/669 (99.55%)
Generation failures	3/669
Valid explanations	492/669
Invalid explanations	177/669
Missing explanations	3
Empty explanations	3

Table 11. Class-wise malware classification report.

Class ID	Precision	Recall	F1-score	Support
0	0.92	0.91	0.91	2196
1	0.77	0.80	0.79	227
2	0.66	0.94	0.78	50
3	0.69	0.52	0.59	386
4	0.80	0.88	0.84	258
5	0.88	0.92	0.90	682
6	0.94	0.94	0.94	2811
7	0.83	0.92	0.87	177
8	0.88	0.94	0.91	1688
9	0.82	0.75	0.78	48
10	0.91	0.71	0.79	314
11	0.81	0.78	0.80	388
12	0.92	0.95	0.94	568
13	0.70	0.70	0.70	895
Accuracy			0.88	
Macro average	0.82	0.83	0.82	10688
Weighted average	0.88	0.88	0.88	10688

Predicted malware family: adware
Observed behavior: displays unwanted advertisements
Potential security risks: data theft, system compromise, unauthorized access, privacy violations
Severity level: high
Malware mitigation actions: isolation of the affected system, perform comprehensive malware scanning, remove malicious artifacts, monitor network communications

4.3 Discussion

The detailed class-wise classification report is presented in Table 11. A confusion matrix is used to visualize the distribution of correct and incorrect predictions, revealing strong diagonal dominance, indicating the majority of samples were correctly classified, thereby minimizing inter-class confusion. Figure 4 provides the confusion matrix for class-wise performance and Figure 5 provides Receiver Operating Characteristic (ROC) analysis, which distinguish malware families, indicating strong separability across different decision thresholds.

Table 12 highlights the key characteristics of malware analysis on CCCS-CIC-AndMal2020 [18] dataset. It presents comparative analysis in terms of malware image and behavioral feature utilization, multimodal fusion, explainability and LLM integration rather than only achieving the highest raw accuracy. Unlike earlier methods, which focus on a single modality [3,4], i.e. malware image representations, and lack behavioral feature integration. Similarly, Vision Transformer (ViT)-based approaches [5] through self-attention mechanisms help in visual-based feature extraction, but are limited to image-based analysis. Multimodal transformers [7] use both visual and behavioral features with an attention-based fusion mechanism; however, to improve model transparency, explainable AI techniques [11] are used. Many LLM models are used in reporting but are not integrated with multimodal malware classification. Our work combines malware image analysis, behavioral feature learning, cross-attention-based multi-modal fusion, explainable AI and LLM threat reporting in a unified model. This helps in high classification results and

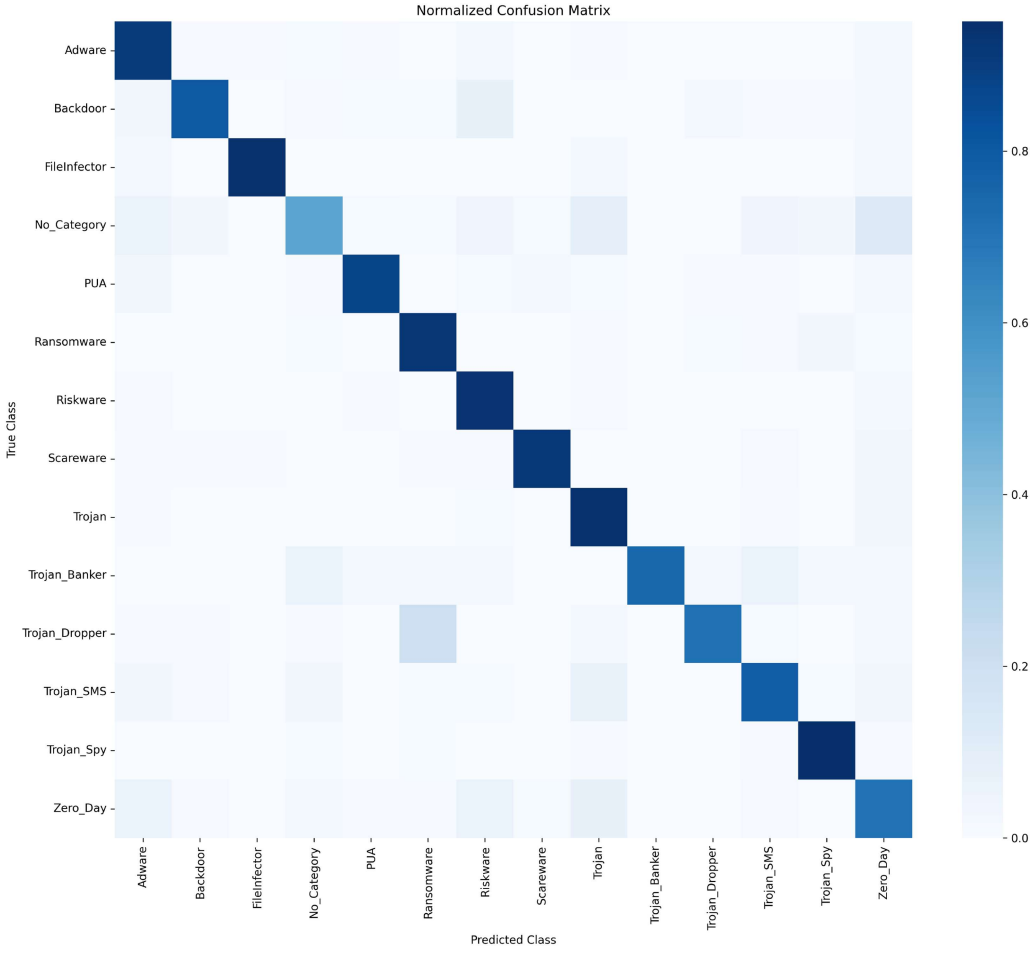


Figure 4. Normalized confusion matrix of the proposed model.

interpretable malware analysis evaluated on the CICAndMal2020 [18] dataset with 88% classification accuracy.

4.4 Ablation study

We quantify our model performance here. When only malware image representations were utilized, the Vision Transformer achieved an accuracy of 81.6%. The model trained exclusively on behavioral features attained an accuracy of 74.8% lacking spatial information. Further, the cross-attention fusion mechanism, obtained an accuracy of 86.3% by explicitly learning interactions between visual and behavioral features. This 6.4 percentage points improvements on over the image-only configuration and 13.2 percentage points improvements on over the behavioral-only model suggest that the optimized fusion effectively combines heterogeneous information sources. Table 13 summarizes the ablation report.

5 Conclusion

Our work presents a multi-model Vision Transformer model that effectively classifies Android malware families by learning malware image representations and behavioral features. It achieved an overall classification accuracy of 88%, with a weighted F1-score of 0.88 and a macro-average F1-score of 0.82 across different malware families. The evaluation results generalizes well across different malware families despite the presence of significant class imbalance in the dataset. Inter-class separability and stable classification performance, malware behavioral attributes and malware report generation together improve the usability of the model for the real-world cybersecurity applications.

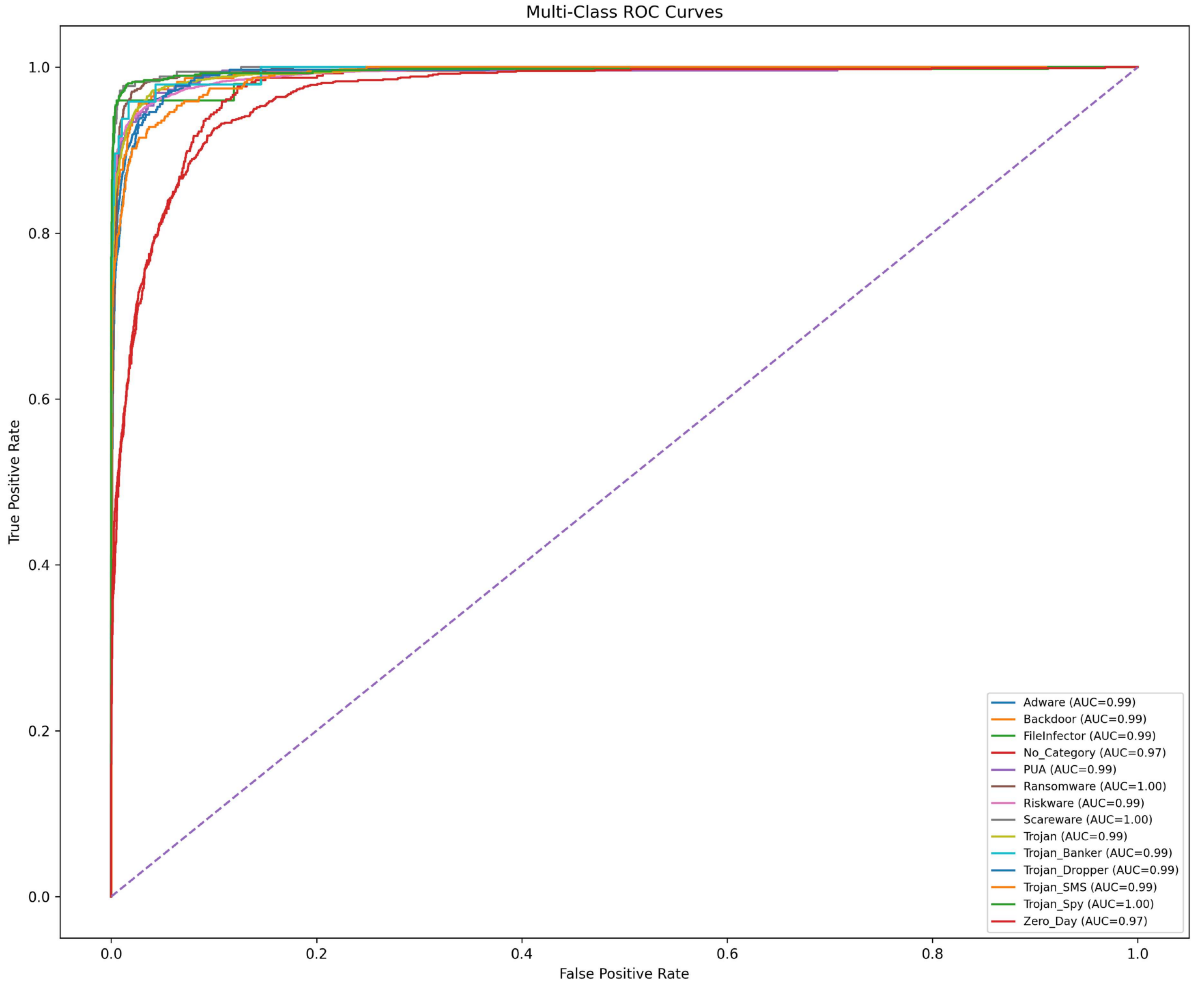


Figure 5. Multi-class ROC curves of the proposed model.

Table 12. Feature level comparison of existing approaches and the proposed work.

Method	Image	Behavior	Cross-attention	XAI	LLM
CNN-based malware classification [4,19]	✓	×	×	×	×
Vision Transformer (ViT) [5]	✓	×	×	×	×
Multi-modal transformer [7,10]	✓	✓	✓	×	×
Explainable artificial intelligence (XAI) [11]	✓	✓	×	✓	×
Large language models (LLMs) [13–16]	×	✓	×	✓	✓
Our work	✓	✓	✓	✓	✓

Table 13. Ablation study.

Model variant	Description	Accuracy (%)
ViT only [5]	Malware image representation using Vision Transformer	81.6
Cross-attention fusion [7]	Cross-attention interaction between visual and behavioral modalities	86.3
Behavioral features only	MLP classifier using behavioral feature vectors	74.8
Proposed multimodal ViT model	Optimized multimodal feature fusion framework	**88.0**

Note: Higher values are in bold.

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Author contributions

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Disclosure statement

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Data availability statement

Dataset used in the experiments is publicly available. Developed models are available on request.

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